PACIFIC DATAVIZ CHALLENGE 2026 · DRAFT FRAMEWORK

The Year We Plan For

Samoa’s rain, heat and the decisions between them.

This framework makes one limited point: resilience cannot be planned around an average year. It shows observed national rainfall variability alongside sea-surface-temperature context — without predicting local water security or claiming causation.

THE QUESTION

What changes when a climate plan is tested against the years Samoa has already experienced?

Country-level and EEZ-level data can show context and variability. They cannot decide what an individual village, catchment or household needs. Local evidence and consent remain an evidence gate.

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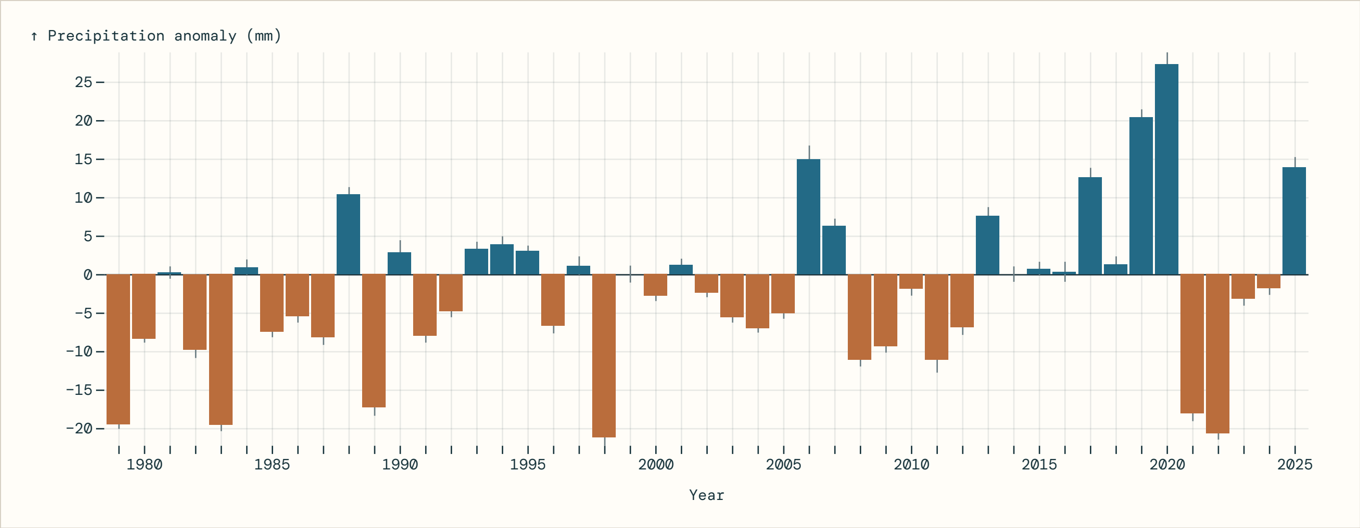
01 · VARIABILITY

The average year is not the year people plan for.

Annual precipitation anomalies, relative to the official 1991–2020 reference period. Bars show measured anomaly; fine lines show reported standard error.



Exploratory annual-data view — not a forecast or causal estimate. Samoa-wide annual series; the inspected metadata does not specify station/grid/national-area aggregation. A negative bar is below the reference average; it is not a local drought declaration.

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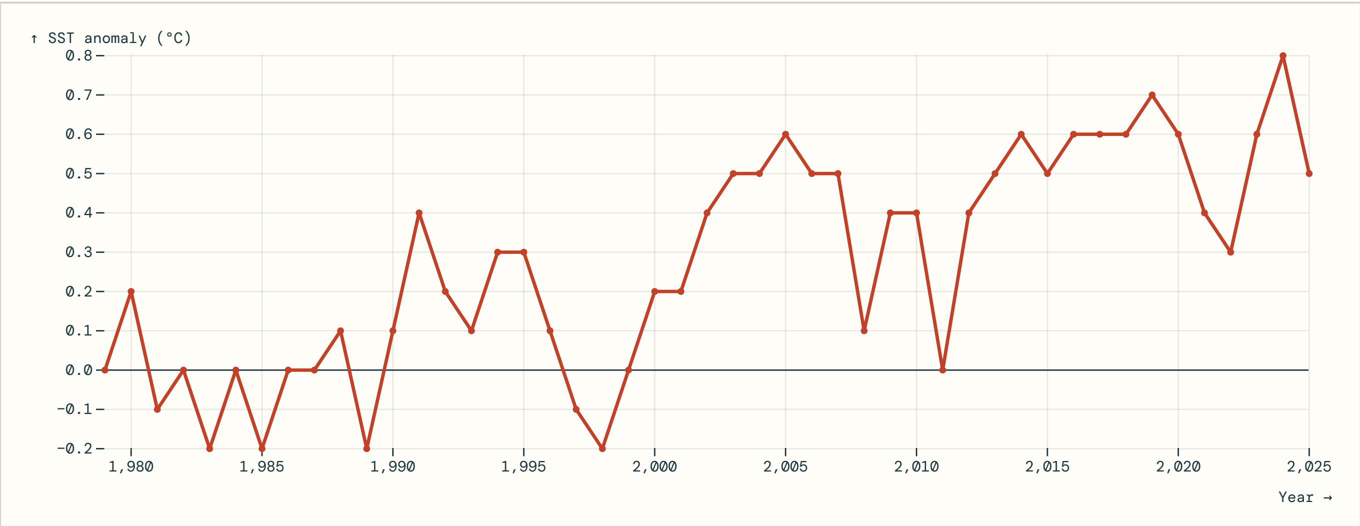
02 · CONTEXT

The ocean backdrop is not static.

Annual Samoa EEZ sea-surface-temperature anomalies, relative to the official 1971–2000 reference period.

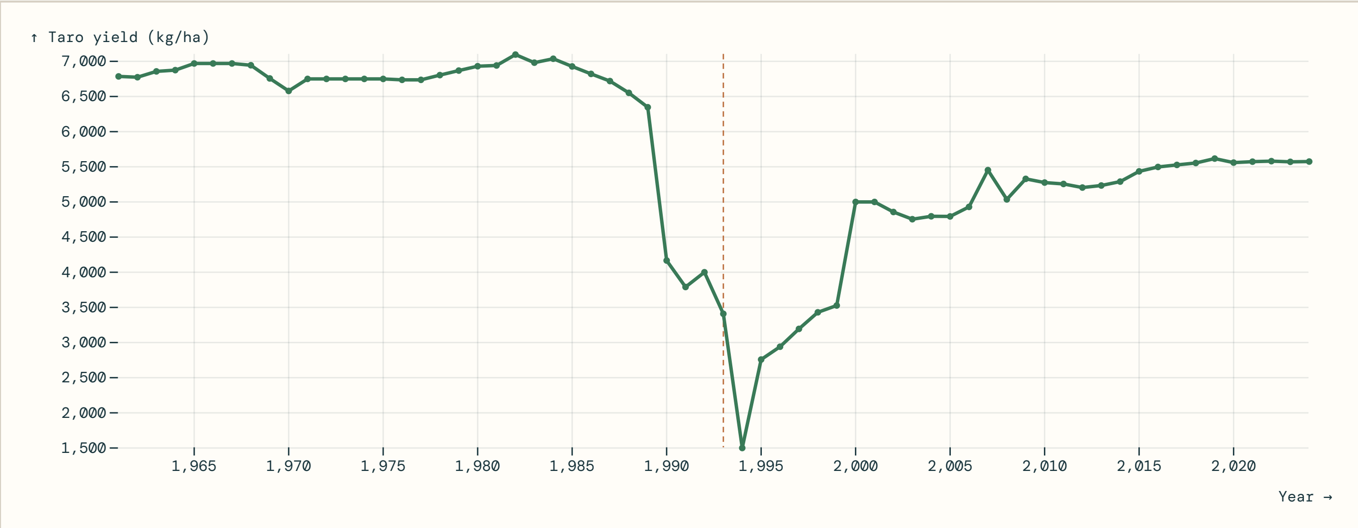


Exploratory annual-data view — not a forecast or causal estimate. This is an EEZ aggregate, not a coastal or inshore temperature record. It is displayed alongside rainfall for context, not as proof of a causal relationship.

03 · FOOD RESILIENCE

Start with taro.

Taro is the most commonly grown root crop and the preferred starchy staple in Samoa, according to SPC's Samoa Agriculture Policy Bank. It is also a strong resilience case: an FAO Samoa study records the destructive taro-leaf-blight outbreak of 1993; SPC later reports disease-resistant varieties increased plantings.



Descriptive national/country-level series — not evidence that climate caused this pattern. Annual taro yield is reported in kg/ha. The sharp 1990s decline is a reason to ask better resilience questions, not permission to attribute outcomes to annual rainfall, SST or any single event. This chart does not forecast yield.

Why this crop: food security, household livelihoods and trade all matter. SPC reports taro was grown by more than 18,347 households in 2015; those figures are historical context, not a current estimate.

Evidence gates for the next build

Seasonal preparedness

Add a below/near/above-normal outlook only when an authoritative product and its hindcast skill are documented.

Taro resilience

Next, add taro production, harvested-area and pest/disease metadata only after their provenance and interpretation are checked. Correlation must not become a causal claim.

Local decisions

Add station, catchment or community detail only with authorised data, quality metadata and, where relevant, local review.

What this draft does not do

- Forecast water-system failure, disaster impacts or crop yields.
- Rank villages or communities.
- Convert missing data into zeroes.
- Use a single number to define "resilience".

Sources and limitations: SPC rainfall SDMX · SPC SST SDMX · SPC taro-yield SDMX · SPC Samoa Agriculture Policy Bank · FAO Samoa taro-leaf-blight history · Challenge data catalogue.